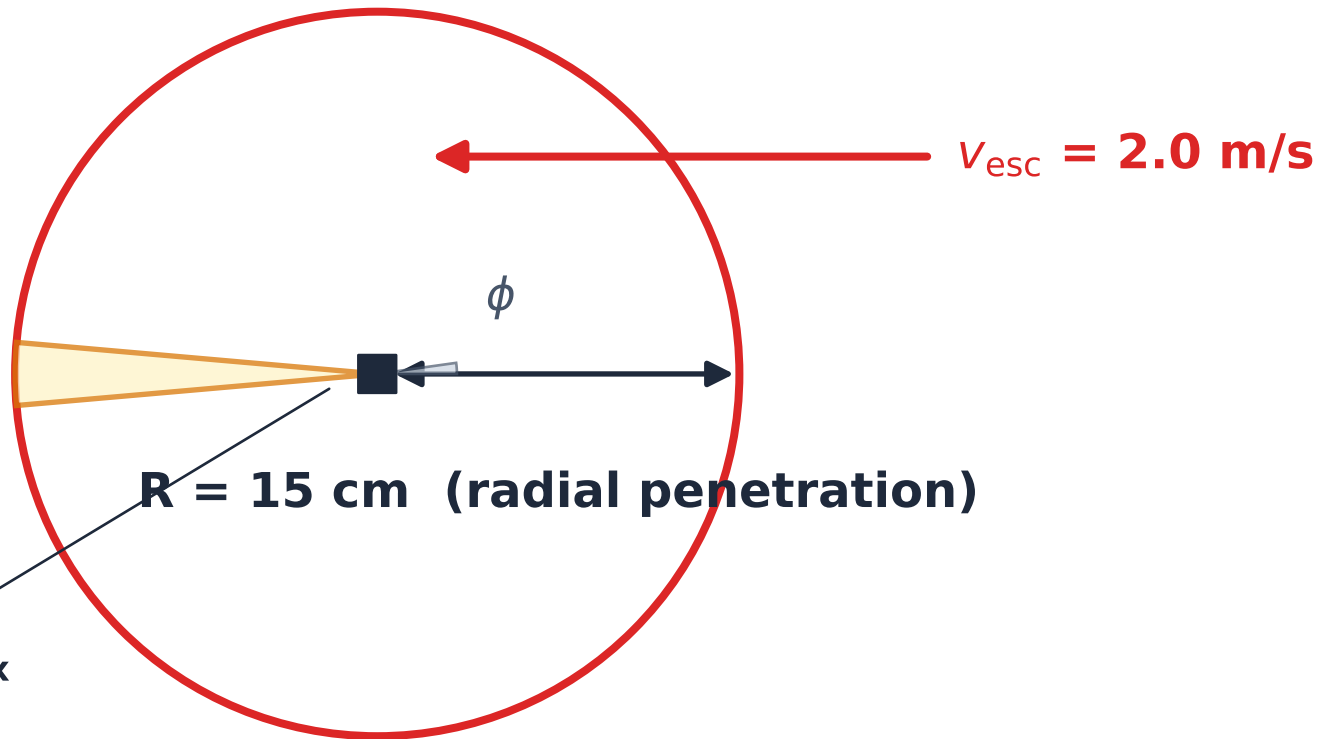


Capture zone (R = 15 cm)



WORST-CASE RADIAL (this work uses this bound)

$$T_{\text{allow}} = R / v_{\text{esc}} = 15 \text{ cm} / 2.0 \text{ m/s} = \mathbf{75.0 \text{ ms}}$$

(strict radial upper bound, $R = 15 \text{ cm}$, $v_{\text{esc}} = 2.0 \text{ m/s}$)

For reference: full diametric traversal $T_{\text{diam}} = 2R / v_{\text{esc}} = 150 \text{ ms}$ (not used in this work)